

5 Interesting Reference Designs For ETHERNET NETWORKS

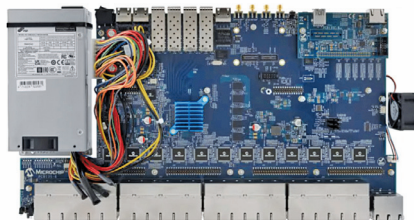
Explore a few Ethernet designs that make building networks faster and easier for data centres, factories, cars, and smart devices.

NIDHI AGARWAL

High-density Ethernet designs help you build networks capable of handling large volumes of data across multiple devices and systems. The reference designs below provide ready-to-use platforms that integrate hardware, software, and connectivity components. Spanning applications from enterprise and data centre switches to automotive gateways, industrial Ethernet, and energy-efficient networks, these designs enable faster, more reliable development, testing, and integration of Ethernet solutions.

High-density Ethernet design

This reference design from Microchip is a complete Ethernet switch system built as a working hardware platform, with the switch chip, PHYs (physical layer transceivers), power, clocking, and management interfaces already integrated, along with supporting



software to run and evaluate the system. It includes a 56-port Ethernet switch with 48 gigabit copper ports, four optical ports, and four multi-gigabit copper ports, enabling deployment in enterprise and data centre environments that require mixed interfaces and high bandwidth.

It supports scalable switching, thermal and power management, and high port density in a compact form factor. Features for monitoring, configuration, and diagnostics simplify integration with existing systems.

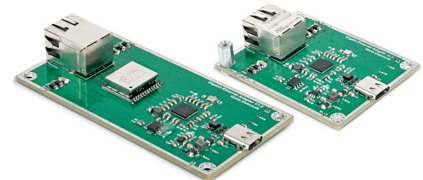
Applications. The reference design is suitable for enterprise backbone networks, aggregation layers, top-of-rack data centre

setups, and industrial or edge networking applications.

Link. <https://www.electronicsforu.com/electronics-projects/health-sensor-platform-4-0-reference-design>
OEM Brand. Microchip

Gigabit Ethernet front-end design

The Gigabit Ethernet front-end reference design from Würth Elektronik provides a framework for building Ethernet interfaces over twisted-pair cables. It includes components such as transformers, common-mode chokes, and capacitors to maintain signal integrity and meet electromagnetic compatibility requirements. Isolation transformers reduce ground-loop issues and improve signal quality, while common-mode chokes limit electromagnetic interference, making the design suitable



for electrically noisy environments. The layout is structured to reduce insertion loss and maintain return loss for high-frequency performance, while surge protection elements protect against voltage spikes.

The design follows IEEE 802.3 standards, ensuring compatibility with existing networks. It provides a strong foundation for designing Ethernet interfaces while simplifying development and integration.

Applications. The reference design can be used across industrial, automotive, and consumer applications, and you can adapt it based on system needs using the provided schematics and layout guidance.

Link. <https://www.electronicsforu.com/electronics-projects/reference-design-for-gigabit-ethernet-front-end>
OEM Brand. Würth Elektronik